AIRFOIL SELECTION FOR TWO RC AIRCRAFT

A simple student-level technical report

Introduction
In this assignment, two RC aircraft are considered. Aircraft A is made to carry more payload and fly steadily. Aircraft B is made for turns, rolls and loops. Since both aircraft have different uses, their airfoils may also be different. This report compares some airfoils and selects one for each aircraft.

1. Assumptions

The assignment does not give the exact size, weight or speed of the aircraft. So, for a simple example, I have assumed air density = 1.225 kg/m³, air viscosity = 1.81 × 10■■ Pa·s, wing chord = 0.18 m and speed = 12 m/s. These are only assumed values. Actual values should be used when the aircraft design is final.

Reynolds number
Re = ρVc/μ
Re = (1.225 × 12 × 0.18)/(1.81 × 10■■) ≈ **146,000**
This value is useful because small RC aircraft usually fly at low Reynolds numbers. Airfoil performance can change at these speeds.

2. Airfoil selection

Aircraft A: High payload and stable flight
The three airfoils considered are Selig S1223, SD7037 and NACA 4412. Selig S1223 is known for producing high lift at low speeds, so it can help an aircraft carry more load. SD7037 is another option for small model aircraft and may give a good balance between lift and drag. NACA 4412 is a common curved airfoil that can produce lift at a small angle of attack.
Selected airfoil: Selig S1223. It is selected because the main aim of Aircraft A is to produce high lift. Its drag and stall behaviour should still be checked before final use.

Aircraft B: Manoeuvrability
The three airfoils considered are NACA 0012, NACA 0015 and Eppler E472. NACA 0012 and NACA 0015 are symmetrical airfoils. This means their upper and lower shapes are similar, which helps during normal and inverted flight. NACA 0015 is thicker, while NACA 0012 is a common balanced option. E472 is another airfoil that can be considered for aerobatic aircraft.
Selected airfoil: NACA 0012. It is selected because it is symmetrical and suitable as a simple starting point for an aerobatic RC aircraft.

3. Comparison of airfoils

Aircraft	Airfoil	Main feature	Use / decision
A	Selig S1223	High lift at low speed	Selected for payload
A	SD7037	Low-speed model-aircraft option	Alternative
A	NACA 4412	Cambered, general-purpose section	Alternative
B	NACA 0012	Symmetrical, 12% thick	Selected for aerobatics
B	NACA 0015	Symmetrical, 15% thick	Alternative; thicker wing
B	Eppler E472	Aerobatic airfoil option	Alternative; needs testing

4. How airfoil shape affects flight

The **camber** is the curve of an airfoil. A more curved airfoil can produce more lift at low speed, which is useful for carrying payload. But it may also produce more drag and is not as suitable for inverted flight. A symmetrical airfoil has similar upper and lower surfaces, so it works more equally when the aircraft is flying upright or upside down.

The **thickness** of an airfoil gives space for the wing structure and spar. A thicker wing can be easier to build strongly, but it may also create more drag. The front part of the airfoil, called the leading edge, also affects how air flows over the wing and how the wing stalls.

5. Can one airfoil be used for both aircraft?

One airfoil can be used for both aircraft if a simple compromise is needed. However, it will not give the best result for both tasks. Aircraft A needs high lift at low speed, while Aircraft B needs similar performance during upright and inverted flight. Therefore, using different airfoils is a better choice for these two missions.

6. Other things that affect aircraft performance

Airfoil is important, but it is not the only thing that decides how an aircraft flies. Other factors include:

- **Wing loading:** the weight supported by each unit of wing area. Lower wing loading generally helps slow flight.
- **Wing shape and size:** wingspan, wing area and shape affect lift and drag.
- **Centre of gravity:** its position affects the stability of the aircraft.
- **Control surfaces:** ailerons and elevators help the aircraft turn, roll and pitch.
- **Weight and structure:** a lighter and strong aircraft can perform better, but it must still be strong enough for its mission.
- **Motor and propeller:** they provide thrust and help the aircraft maintain speed.

7. Basic lift equation

For steady level flight, lift is approximately equal to the aircraft's weight.

$$L = \tfrac{1}{2}\rho V^2 S C_L$$

Here, ρ is air density, V is speed, S is wing area and C_L is lift coefficient. This equation can be used with the actual aircraft weight and wing area to check whether the wing can produce enough lift.

8. Conclusion

For Aircraft A, Selig S1223 is selected because it can produce high lift at low speed, which is useful for carrying payload. SD7037 can be considered if a better balance between lift and drag is needed. For Aircraft B, NACA 0012 is selected because its symmetrical shape is useful for aerobatic flying, including inverted flight. The final choice should be checked by testing the airfoils at the actual flight speed and wing size.

References

[1] UIUC Airfoil Coordinates Database, University of Illinois: https://m-selig.web.engr.illinois.edu/ads/coord_database.html

[2] M. Drela and H. Youngren, *XFOIL 6.9 User Primer*, 2001: https://v0xnihili.github.io/xfoil-docs/

[3] NASA Glenn Research Center, *Inclination Effects on Drag*: https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/inclination-effects-on-drag/

Note: The airfoil comparison in this report is based on general characteristics. XFLR5/XFOIL graphs have not been added because no simulation has been performed. For a complete bonus analysis, the airfoils should be tested at the same Reynolds number and the graphs should be included.